

Democratizing ecological data: An open-source toolkit for connecting tracking and remote sensing data in migratory bird systems

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Summary

Since the advent of satellite remote sensing and animal telemetry, conservation research has increasingly drawn on spatial and movement ecology to understand species–habitat interactions. However, the ability to engage fully with these data depends on having accessible tools and workflows that match the diverse skills and capacities of practitioners and ecologists. Addressing this gap requires the availability of user-friendly and accessible data processing pipelines and interfaces. Moreover, leveraging such knowledge to support management decisions requires straightforward access for conservation practitioners and decision-makers to visualize and explore meaningful outputs. The need for (online) toolkits is particularly pronounced in the case of transboundary research and conservation of migratory birds. An exemplary case is the growing portfolio of local and flyway-scale collaborations for the conservation of continental Black-tailed Godwits (*Limosa limosa limosa*) in the African-European flyways. To facilitate the exploration and analysis of large ecological datasets gathered through a multitude of parties and projects, we developed a toolkit including workflow and coding pipelines, and a suite of interactive web apps. This toolkit combines Google Earth Engine (GEE) and R Shiny Apps to allow stakeholders to integrate telemetry data with remote sensing-derived environmental layers across the godwit's full spatial-temporal migratory range. The GEE Apps, designed for broad applicability across ecological study systems, provide on-the-fly access to remote sensing products, including grassland productivity, seasonal surface water extent, and land cover classification, without requiring users to perform complex data preprocessing. Building on these outputs, the R Shiny Apps connect processed environmental layers with godwit movement data. Users can (1) generate summary statistics regarding godwit tracking data availability within custom timeframes and areas of interest, (2) examine breeding habitat use in relation to grassland productivity at one of their breeding strongholds in Northwest Europe, (3) map seasonal surface water availability at a key stopover site in Iberia, and (4) evaluate habitat use during the non-breeding period in a West African residency site. By effectively supporting users in the exploration of spatial and movement data, this toolkit is a step towards fostering more inclusive evidence-based conservation decision-making for threatened migratory species navigating dynamic, fragmented, and connected landscapes pressured by global changes.

Keywords: Google Earth Engine, R Shiny, telemetry, godwit, satellite, Landsat, Sentinel, practitioners, East Atlantic Flyway, web apps

1. Introduction

Migratory birds are among the species that illustrate the global challenges for biodiversity conservation (Newton 2004; Wilcove & Wikelski, 2008; Rosenberg et al., 2019). As their life cycle involves navigating a patchwork of habitats that vary rapidly across space and time, their survival and fitness depends not only on the habitat quality of their breeding grounds

but also on the conditions encountered in migratory flight, during refueling at staging sites, and at non-breeding residency sites (Marra et al., 2015; Rakhimberdiev et al., 2019). Cumulative pressures that build across the full annual cycle can turn otherwise isolated stressors into system-wide bottlenecks (Buehler & Piersma 2008). These bottlenecks emerge when local threats, such as habitat loss or climatic extremes, are compounded across multiple stages of the migratory journey (Newton et al., 2006; Piersma et al., 2016; Conklin et al., 2021). As a result, conserving migratory birds requires not only protecting key habitats, but also a spatially explicit understanding of where, when, and how stressors interact across the flyway (Runge et al., 2015).

In Europe, migratory meadow birds like the Eurasian Curlew (*Numenius arquata*), Ruff (*Calidris pugnax*), Lapwing (*Vanellus vanellus*), and Black-tailed Godwit (*Limosa limosa*) depend on diverse, structurally complex grassland landscapes, that constitute their habitats during the breeding period (Thorup, 1997). But monocultures formed by high nutrient inputs, deep and rapid drainage, and frequent mowing are eroding the ecological value and resilience of these landscapes (Harvey, 1997). These pressures have driven widespread declines in farmland biodiversity across Europe, prompting policy responses such as agri-environmental measures under the Common Agricultural Policy (Pe'er et al., 2014). Meanwhile, throughout their non-breeding range, expanding irrigation, prolonged drought, and land conversion are contributing to rapid loss and degradation of key wetland habitats (Davidson, 2014; Navedo et al., 2022). Nevertheless, the knowledge required to support the most effective conservation actions and policy is challenging to obtain (Sutherland et al., 2004), closely related to adoption of technologies (Eren et al., 2024) and depends on data sources that are not easy to measure, process and interpret. Building this knowledge depends on ecological research that directly, or indirectly, supports management, grounded in understanding habitat use across life stages and how environmental pressures accumulate spatially (Marra et al., 2015). Spatial ecology provides a critical framework for linking animal movements to the physical characteristics of their environment. By integrating habitat data with movement data, researchers can examine how changes in environmental conditions are associated with changes in animal behavior at multiple spatial-temporal scales, such as shifts in movement patterns or space use across time (Matthiopoulos et al., 2023).

In the field of spatial ecology, satellite remote sensing has greatly advanced our ability to monitor landscapes, offering near-continuous coverage that is both near-global in scope and fine in resolution (Pettorelli et al., 2014). These data are now widely used to detect landscape-level changes (Muchoney et al. 2000; Howison et al. 2018), map biodiversity (Turner et al., 2003), and facilitate collaboration across disciplines and institutions (Robbins, 2003; León, 2022; Garzon Lopez et al., 2024). However, the expertise needed to acquire, process, analyze, and interpret remote sensing data is often limiting for many conservation practitioners and researchers (Wegmann, 2017). The very characteristics that make remote sensing so powerful – its scale, volume, and richness – also make it challenging to apply. Selecting appropriate sensors, balancing spatial and temporal resolution, and processing raw imagery all require navigating a multitude of technical decisions that can influence outcomes (Atkinson & Curran, 1997). Beyond image selection, optimizing coverage and maintaining image quality is no small feat. Atmospheric factors such as cloud cover and haze, along with technical issues like sensor malfunctions (e.g. Scan Line Failure of Landsat 7; Sentinel-1B failure) can result in incomplete time series. To mitigate these problems, analysts use cloud masking techniques and rigorous quality assurance protocols to filter out unreliable data (Gorelick et al., 2017). In some cases, data

fusion or gap-filling algorithms are used to combine information from multiple sources (Zhu et al., 2018; Henriques et al., 2022). All these technical considerations can constitute an additional hurdle for ecologists and conservation researchers, since they require particular skill sets and considerable time investment that might not be compatible with the priorities of these stakeholders. By alleviating such challenges, our aim is to contribute to their capacity to leverage remote sensing tools to advance ecological knowledge for conservation.

Analogous to how satellite imaging has allowed for ‘remote sensing’ of habitat changes, tracking and biologging technologies allow us to remotely measure the movements, and thus the use of distant environments, by migratory birds (Gregory et al., 2023). Tracking bird movements using loggers and transmitters not only helps identifying migratory routes and site selection patterns (Chan et al., 2019), but also provides a potential way to connect ecological research with practical conservation goals, such as identifying priority sites through movement-based connectivity analyses of key habitats (Beal et al., 2025). While maps can mislead (Monmonier, 1996), visualizing animal location points through maps also taps into our natural sense of space, making animal movement broadly interpretable by researchers, decision-makers, and the public alike. Technological advancements in the last decade meanwhile enabled miniaturization of telemetry devices, such that tracking can now contribute to conservation research on a much greater number of species (Kays et al., 2015). However, as with remote sensing data, animal telemetry data are challenging to work with (Henriques et al., 2025). Unlike satellite imagery, which is organized in neat, gridded formats, telemetry data are often collected at irregular intervals and with varying spatial accuracy (Fleming et al., 2018). These inconsistencies make it difficult to align movement records with environmental datasets, particularly when trying to extract habitat conditions at specific times and places. Moreover, effective use of telemetry data depends on well-maintained metadata that captures details such as device type, tag deployment history, and individual bird identity and traits (Michel et al., preprint).

Integrating remote sensing and telemetry datasets also involves overcoming practical limitations, such as computational hurdles and financial costs. High-performance computing and large-scale data storage are often essential but expensive, creating barriers for local research, especially in low-income regions (De Vega et al., 2019). These financial constraints can severely limit access to the necessary hardware, cloud services, or even reliable internet infrastructure for processing large datasets. Ironically, these challenges often disproportionately affect areas where biodiversity is richest and conservation efforts are urgent (Amano & Sutherland, 2013; Jenkins et al., 2013).

We live in an era defined not by a lack of data, but by an overabundance of it – especially in ecology, where some aspects of research are increasingly data-driven (Hampton et al., 2013). However, handling this vast amount of information responsibly and to produce the knowledge we still lack, presents additional difficulties (Pritchard et al., 2022), including the need for adequate indicators to process the data into an actionable form (Affinito et al., 2024), biases in sampling (Hughes et al., 2021; Amano et al., 2023) as well as the complexity of dealing with different data types and sources (Xia et al., 2020). In an inherently interdisciplinary field such as conservation ecology, researchers typically assume multiple roles, ranging from field biologists to statisticians and data managers – often without training (Barraquand et al., 2014). Rapid advances in data collection and analysis tools only add to this burden (Hampton et al., 2017). While the technical barriers can be steep, they have also spurred a wave of innovation, like R Shiny apps (Chang et al., 2025) for visualizing and assessing site connectivity along the East Atlantic Flyway (Beal et al. 2025), Google Earth Engine (GEE) apps (Gorelick et al., 2017) for reconstructing historical wetland

dynamics in Doñana (de Felipe et al., 2024), and the MoveApps platform (Kölzsch et al., 2022), which enables cloud-based analysis modules built on top of Movebank's infrastructure (Kays et al., 2022). Meanwhile, initiatives like Global Wader (<https://www.globalwader.org/>), aim to become a central register for wader tracking projects around the world, facilitating collaboration between data owners and users. Efforts like these demonstrate not only the feasibility but also the growing momentum behind open-access tools that make spatial data more transparent and accessible. Additionally, by making the underlying workflows openly accessible, quantitative and spatial ecologists can adapt, replicate, or extend existing methods to fit new questions or regions. This openness enables a building-block approach where each contribution becomes part of a shared foundation that others can refine and expand.

In this paper, we introduce a remote sensing and movement data toolkit designed to support and enable stakeholders working on migratory meadow bird systems. This toolkit is intended for two core audiences. The first group consists of researchers focused on meadow bird ecology, including those studying bird behavior, food web dynamics, landscape interactions, and the effects of land use intensity or climate change on meadow bird ecosystems. These stakeholders require spatial ecological data to inform their work and may benefit from tools that provide insight into the availability and biases in tracking data, and guidance on working with remote sensing and movement analyses workflows. The second audience is conservation professionals and managers working directly with migratory meadow bird systems – those responsible for monitoring species and their habitats, implementing conservation actions and management decisions, and responding to real-time changes in these systems. These stakeholders often face pressing decisions that could be supported by large spatial and movement datasets, if the tools to interpret these were more readily available.

1.1. Case study: the Black-tailed Godwit system

To illustrate the application of the toolkit, this paper focuses on the continental Black-tailed Godwit (*Limosa limosa limosa*), hereafter referred to as godwit – a flagship migratory meadow-breeding waterbird whose ecological and conservation context represents many of the broader challenges faced by meadow-breeding species dependent on a network of wetland habitats across flyway scales. Birds of the *limosa* subspecies migrate along the East Atlantic Flyway, depending on a set of sites for different life-cycle moments that range from North Europe through the West African region (Hooijmeijer et al., 2014; Verhoeven et al., 2021). They typically arrive at their breeding grounds in Northwest Europe in late March to April, where they nest and rear chicks through June (Kentie et al., 2018). Post-breeding migration begins as early as mid-June, with many individuals stopping over in wetland sites in France and the Iberian Peninsula. By August, most godwits reach their non-breeding residencies, which include both West Africa and parts of southern Iberia. They remain at their non-breeding residencies until the migration back to the breeding grounds begins, often making prolonged stopovers in southern Europe along the way (Masero et al., 2017; Verhoeven et al., 2018; Beal et al., 2025).

Godwits breed in the low-lying wet grasslands of northwestern Europe during the northern spring season (March – June), with the core breeding populations found in the Netherlands and Germany (Thorup, 1997). These breeding landscapes were historically maintained through low intensity grazing and mowing, and represented a relatively heterogeneous wet grassland system. Most of these areas are now dominated by intensive dairy farming systems (Howison et al., 2018; Barba-Escoto et al., 2024). In the Dutch

province of Friesland and the German state of Lower Saxony, some of the most important strongholds for breeding godwits, grassland productivity is tightly managed through precise combinations of fertilization, soil drainage, irrigation, and mowing (Groen et al., 2012). This has produced homogenous, high-output grasslands where fast-growing ryegrasses dominate and compacted soils limit the activity of soil fauna (Onrust et al., 2019). Draining soils serves dual purposes: enabling access for heavy machinery and weed control, both central to maintaining fast-growing monocultures of ryegrass. The same practices that enhance short-term productivity have eroded the system's resilience, leaving it increasingly vulnerable to climatic extremes such as drought and heat stress (Joyce et al., 2016; Kraamwinkel et al., 2025).

During pre-breeding migration, godwits depend on several key stopover sites across southern Europe to refuel *en route* to their breeding grounds (Hooijmeijer et al., 2014). One such site is the Doñana wetlands in southern Spain, where recent years have seen major hydrological shifts due to drought and groundwater overextraction for agriculture (Green et al., 2024; Vansteelant et al., in prep). In addition, the Tagus estuary in Portugal, along with its surrounding rice fields, has supported an increasing number of godwits in recent years (Verhoeven et al., 2018). As godwits rely on the estuary's saltpans and rice fields to refuel during this critical phase of migration (Lourenço et al., 2010), the hydrological management and water-table regimes of the Tagus, as well as other water-stressed sites across the Mediterranean, are of particular interest to researchers and projects focused on the godwit system, such as the LIFE Godwit Flyway (2023) project.

Further south, many godwits spend the non-breeding residency in West Africa, including the Senegal River Delta, in West Africa, where they use ephemeral wetlands such as floodplains and rice fields shaped by dam-regulated hydrological regimes (Craft et al., 2024). The delta represents one of several key non-breeding areas across West Africa (Hooijmeijer et al., 2014; Howison et al., 2019; Verhoeven et al., 2021), and like other sites in the region, faces growing pressures from altered hydrology and agricultural expansion (Mietton et al., 2007).

We specifically focus on key areas used by godwits of the *limosa* subspecies, tagged across multiple research projects in several countries. These tagged individuals represent the majority of the movement data archived in Movebank for this subspecies along the East Atlantic Flyway.

1.2. Toolkit overview

To explore migratory bird movements across such diverse and dynamic landscapes, researchers and conservation professionals need accessible tools that link spatial environmental data to the full annual cycle of the birds. The Godwit Landscapes toolkit introduced here meets this need through two complementary types of web applications, each tailored to different needs and use cases: GEE apps, and R Shiny Apps (see below). Together, these include seven applications: three GEE apps and four R Shiny apps. The GEE apps provide on-demand processing and access to remote sensing layers used across the toolkit, while the R Shiny apps are designed to visualize godwit movements in relation to specific habitat features or landscape conditions, building on the GEE layers produced with the GEE apps workflow. Table 1 summarizes each app's platform, primary function, and output type.

Google Earth Engine Apps

These apps are intended for researchers and field ecologists who need access to remotely

sensed data as environmental layers of the study system without having to engage in extensive data preprocessing, coding, and with other technical details needed to produce these data layers. Users can visualize, generate, and export spatial raster datasets directly within a browser-based interface, with no programming required. A moderate familiarity with raster data handling is required for further interacting and interpreting these outputs, while users requiring more tailored outputs can access the underlying GEE Code Editor scripts to customize workflows or fine-tune export parameters.

R Shiny Apps

The processed raster outputs from the GEE apps are integrated into standalone R Shiny applications. While the GEE apps are broadly applicable across ecological systems, the R Shiny apps are designed specifically around our case study, the godwit system, focusing on three key sites along their flyway: the southwest Friesland region, the Doñana wetlands, and the Senegal River Delta system. These apps are tailored to conservation professionals, including nature managers and policy makers, and applied ecologists who need intuitive tools for contextualizing godwit movement with remote sensing variables of their habitat across their flyway and annual cycle. No coding is required, and all analyses are performed at the back-end, prompted through user-friendly selection menus and interactive maps. These allow users to visualize and interpret near real-time habitat and godwit movement data at a flyway scale.

Table 1. Descriptive overview of the three GEE Apps and the four R Shiny Apps that make up the toolkit.

App Name	Platform	Primary Function	Main Outputs
Grassland Production Intensity (GPI) Tool	GEE	Mapping of grassland production intensity	Custom-made grassland production intensity raster based on Sentinel 2 satellite imagery (GeoTIFF: 20 m)
Seasonal Surface Water Mapping Tool	GEE	Mapping of seasonal surface flooding of wetlands	Custom-made flood raster (GeoTIFF: 30 m) based on Landsat satellite series
Land Cover Classification Tool	GEE	Personalized supervised classification of land cover types	Land cover/habitat raster based on Landsat satellite series (GeoTIFF: 30 m)
Flyway Movement Viewer	R Shiny	Visual exploration of flyway-scale godwit movements across annual cycles; Compute summaries of tracking data availability	Movement trajectories; movement and tracking data summary statistics within default and user-defined areas of interest across time-series
Grassland Production Intensity Viewer in Southwest Friesland	R Shiny	Visual exploration of godwit space use in relation to grassland production intensity in a key Dutch breeding area	Interactive map of godwit locations overlaying a GPI map; histogram of godwit locations across GPI values
Doñana Wetland Seasonal Flooding Viewer	R Shiny	Visual exploration of godwit space use in relation to seasonal drought conditions	Interactive map of godwit locations overlaying a flooding map; bar chart quantifying godwit locations in flooded VS dry pixels

Senegal Delta Habitat Use Viewer	R Shiny	Visual exploration of godwit non-breeding habitat use patterns in relation to climatic seasons at a key non-breeding residency site	Interactive map of godwit locations overlaying Land use/habitat classes; Bar chart quantifying godwit locations in each habitat class in dry VS wet season
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2. Methods

2.1. Data sources and processing pipelines

2.1.1 Telemetry data sources and processing

The Godwit Landscapes toolkit relies on a long-term dataset of godwit GPS and Argos telemetry, compiled from numerous Movebank studies (see Section 5.4 for an overview). To ensure the toolkit reflects up-to-date bird movement information, tracking data are retrieved weekly from Movebank (Kays et al., 2022) through an automated execution of a scheduled R script built around the *move* package (<https://cran.r-project.org/package=move>), which retrieves and merges tracking data from nearly 40 different Movebank godwit studies. After downloading, we implemented a quality control routine that removes implausible locations using a rolling mean and standard deviation calculated over a window of five-locations (latitude-longitude coordinate pairs). We used these rolling statistics to compute a standardized squared deviation for each location, flagging those exceeding a value of 10 (approximately equivalent to 3.2 standard deviations) as outliers. Furthermore, only Argos location classes 1, 2, or 3 were retained, which are associated with estimated error radii below 1500 meters. Lower-quality classes (0, A, B, C, and Z), which either exceed this threshold or lack a defined accuracy estimate, are excluded. The error radius for each Argos location is provided in the metadata and reflects a 68% confidence interval for positional accuracy, based on satellite geometry, the number of transmissions received, and Doppler shift calculations (Vincent et al., 2002).

2.1.2. Remote sensing data layers

The toolkit integrates three remote sensing-derived environmental layers to support analyses of space and habitat use and other ecological research on the godwit system, namely the grassland production intensity, surface water extent, and land cover classification. All datasets were generated using freely available satellite imagery and processed within cloud-based workflows in GEE. Each product is designed to align spatially and temporally with godwit tracking data, supporting habitat assessments across multiple regions and phases of the annual cycle. These remote sensing layers form the foundation of both the GEE apps and the R Shiny applications.

Grassland production intensity

The grassland production intensity (GPI) layer serves as a proxy for aboveground biomass in agricultural grasslands. This product uses the Sentinel-2 Red Edge Position index (S2REP), a spectral metric sensitive to canopy chlorophyll concentration and vegetation structure (Delegido et al., 2011; Schlemmer et al., 2013), making it well-suited for assessing intensively managed dairy pastures (Bekkema & Eleveld, 2018). S2REP is calculated from Sentinel-2 satellite spectral bands B4 (red), B5 (red edge 1), B6 (red edge 2), and B7 (red edge 3) using the formula:

$$S2REP = 705 + 35 \times ((B7 + B4) / 2 - B5) / (B6 - B5))$$

In practical terms, S2REP values differentiate fields based on their biomass – which most often is a result of management practices. Grasslands with higher S2REP values typically exhibit denser, more chlorophyll-rich vegetation, which is often a sign of more intensive management (Bekkema & Eleveld, 2018). In contrast, lower values suggest less productive or less intensively managed areas. Because mowing reduces the biomass – consequently affecting the spectral reflectance of grasslands, it is important to capture S2REP imagery on a consistent date (typically once per year or growing season) before the onset of the mowing season. This approach ensures that biomass differences between fields are comparable and not confounded by variations in mowing dates.

Seasonal surface flooding

The seasonal surface flooding layer estimates the surface water extent based on shortwave infrared (SWIR) reflectance from Landsat imagery. SWIR wavelengths are highly sensitive to the presence of water due to strong absorption by liquid water molecules in this part of the electromagnetic spectrum (Gao, 1996). As a result, surface water bodies reflect much less SWIR radiation compared to dry soils or vegetation, making SWIR an effective wavelength for detecting flooded areas (Li et al., 2015; Lefebvre et al., 2019). The SWIR band is available in multiple Landsat satellites and as it is a single band, requires relatively minimal preprocessing. This is favourable for supporting consistent, long-term monitoring of hydrological changes at local to global scales (Wulder et al., 2019), including wetland habitats (Díaz-Delgado et al., 2016).

Land cover classification

In this toolkit, land cover classification is performed using imagery from Landsat 8 and 9, the latest satellites in the Landsat program. These satellites capture 30-meter multispectral imagery across visible, near-infrared (NIR), and shortwave infrared (SWIR) bands. We use Landsat's Collection 2 Level-2 dataset (Earth Resources Observation and Science (EROS) Center, 2020), which provides atmospherically corrected imagery with quality assurance (QA) bands used to mask clouds, shadows, and radiometric saturation. The underlying principle is that different land cover types reflect light differently across the electromagnetic spectrum, producing unique spectral signatures. By collecting a set of georeferenced training points from a known date (as seasonality can affect classification) and associating each point with a known class label, a machine learning algorithm can be trained through a supervised classification process, to learn these spectral distinctions (Jensen, 2005). Training points may be sourced from field surveys, existing maps, or satellite imagery where land-cover classes are clearly distinguishable. Proper timing is site-dependent and requires awareness of landscape dynamics with enough prior knowledge to align sampling with class conditions.

Spectral reflectance values from different bands, typically in the blue, green, red, near-infrared (NIR), and shortwave infrared (SWIR) ranges, are used as input features for classification. In addition to these raw spectral bands, vegetation indices such as the Normalized Difference Vegetation Index (NDVI) are often included to increase class separability. NDVI, which is calculated from red and NIR bands, is sensitive to vegetation greenness and biomass and is widely used to differentiate vegetated from non-vegetated surfaces (Sha and Yu, 2008; Pettorelli, 2013).

2.1.3. Automation pipeline

The GEE apps are built to fetch and process satellite imagery dynamically based on user input, requiring no manual updates. In contrast, the R Shiny apps rely on preprocessed raster layers that must be periodically updated by the authors for the latest imagery. This update process involves downloading the latest GEE-derived rasters, performing visual checks for quality control, and adding the validated layers to a centralized GitHub repository. In parallel, a GitHub Actions workflow – a cloud-based automation system integrated into the GitHub platform – is used to automatically run scripts for telemetry data processing and R Shiny App publication. This workflow is triggered on a fixed schedule or when new code is pushed to the repository, allowing data to be updated and apps to be redeployed without manual intervention. This pipeline retrieves new movement records from Movebank each week, applies filtering and subsampling routines (see 2.1.1.), and pushes the updated dataset to the GitHub repository. From there, updates are automatically deployed to shinyapps.io, a cloud hosting service for R Shiny applications that allows users to run interactive apps directly in a web browser without needing to install R or any dependencies. This workflow, presented in Figure 1, removes the need for manual updates, streamlines app maintenance, and enables near real-time data availability for end-users.

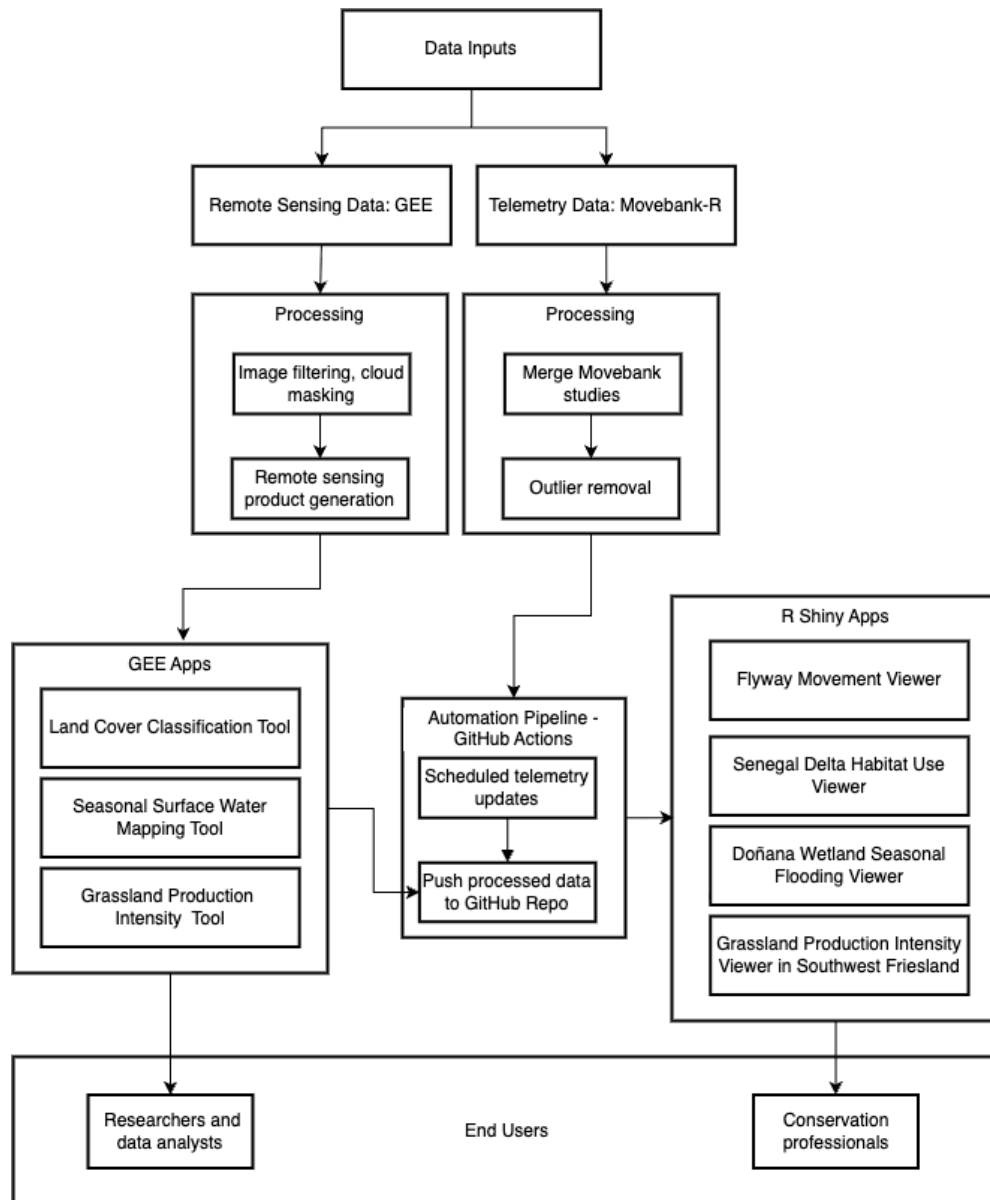


Figure 1. Automated data processing and deployment workflow for the Godwit Landscapes Toolkit.

2.2. Application design

2.2.1. Google Earth Engine Apps

Each GEE App includes two components: a backend raster generation script, developed in the GEE Code Editor, and a simplified graphical user interface (GUI) deployed as a public-facing GEE App. The Code Editor is a JavaScript-based development environment that provides access to the full cloud-based geospatial processing capabilities of GEE. This environment allows technical users to define workflows for imagery filtering, compositing, masking, and export, using custom logic and advanced geospatial operations. These script-based workflows are best suited for users with programming experience who need precise control over data inputs, analyses parameters, and export settings. In parallel, the GEE apps include a web-based GUI designed for accessibility and ease of use. These interfaces are built using GEE's user interface (UI) module, which supports the interactive app construction. GEE app interfaces typically include dropdown menus for selecting dates,

regions, or variables; checkboxes for toggling layers; and interactive map tools for drawing points or polygons. Once deployed, the GUI is hosted on Google's infrastructure and can be accessed through a public URL, allowing users to run complex remote sensing workflows without registration or coding.

Grassland Production Intensity Tool

To generate the GPI layers, we accessed Sentinel-2 Level 2A Surface Reflectance imagery (COPERNICUS/S2_SR; European Space Agency, 2021) through GEE. Scenes were filtered using a 65% threshold from the COPERNICUS/S2_CLOUD_PROBABILITY dataset to remove high-cloud cover images. Additional spectral edge masking was applied using Sentinel-2 bands B8A and B9 to exclude scenes affected by haze or residual atmospheric interference. For each year, the least-cloudy image was identified based on the CLOUDY_PIXEL_PERCENTAGE property. We computed the S2REP index from each selected image and retained the resulting S2REP band. The derived raster was then exported as a single-band GeoTIFF at 20-meter spatial resolution.

The GEE code editor version enables technical users to define AOIs, set timeframes, and export GPI raster with full control over filtering and processing steps. To support broader accessibility, the same logic was translated into a public GEE App using the platform's UI components. This app guides users through selecting a date range and location, choosing from the least-cloudy scenes, and defining a custom clipping boundary for export. Both workflows use identical backend operations, ensuring consistency across outputs regardless of user experience or interface preference.

Seasonal Surface Water Mapping Tool

To generate the surface water composites, we used Tier 1 Level-2 surface reflectance imagery from Landsat 7, 8, and 9, accessed through GEE (collections: LANDSAT/LE07/C02/T1_L2, LC08/C02/T1_L2, and LC09/C02/T1_L2; United States Geological Survey, 2021). To ensure consistency across Landsat satellites, band names were harmonized by renaming Landsat 7 bands to match the naming conventions of Landsat 8 and 9. All images were scaled to surface reflectance using standard coefficients, and quality assurance was applied using the QA_PIXEL band. Bitmask filters were used to remove clouds and cloud shadows, ensuring clean observations across all three sensors. From each collection, we extracted the SWIR1 band (band 6 for Landsat 8/9 and band 5 for Landsat 7). After preprocessing, images from all sensors were merged and composited into a single monthly median image for each user-defined date and area. These composites were clipped to a 30 km buffer surrounding a user-selected point and exported as single-band GeoTIFFs with 30-meter spatial resolution.

The interactive version of the tool was implemented as a GEE App using Earth Engine's UI framework. The app reproduces the same backend logic within a graphical interface, allowing users to define an Area of Interest (AOI) by clicking on the map and specifying a year and month using dropdown menus. The app automatically filters and processes the imagery, displays the SWIR-based composite on the map using a blue-to-grey color palette, and provides a direct download link to export the clipped raster for offline analysis.

Land Cover Classification Tool

Similar to the previous GEE apps, two parallel implementations were developed for supervised land cover classification: a code-based workflow for technical users and an

interactive GEE App. The code-editor version is structured to walk users through the core steps of classification. Users define an AOI by drawing, importing, or defining the geographic coordinates of a polygon. Landsat 8 and 9 imagery from the Level-2 surface reflectance collection (LANDSAT/LC08/C02/T1_L2, LANDSAT/LC09/C02/T1_L2; United States Geological Survey, 2021) is filtered by date and clipped to the AOI. To ensure spectral consistency and reduce noise, each image is atmospherically corrected and masked for clouds, shadows, and saturated pixels using QA_PIXEL and QA_RADSAT bands. A median composite is then generated from the filtered image collection, producing a cloud-free base layer for classification.

Training data are manually created by placing labeled points within the AOI, each assigned a land cover class represented as an integer. These points are merged into a single feature collection and used to sample spectral values from the composite image. A Random Forest (Breiman et al., 2001) classifier with 500 decision trees, as suggested by Belgiu & Drăguț (2016), is then trained on this labeled dataset and applied to the full image. The result is a spatially explicit land cover raster with user-defined classes. Classification accuracy can be optionally assessed by splitting the training data into training and validation subsets and calculating a confusion matrix. The final classified raster can be exported to Google Drive as a 30-meter resolution GeoTIFF using GEE's `Export.image.toDrive()` function.

To support broader audiences, the same workflow was adapted into a step-by-step GEE App using the Earth Engine UI module. The app presents a sequential interface guiding users through year and month selection, training point collection, classification execution, and export. Users begin by selecting a date range and base imagery from Landsat 8 and 9 (filtered for low cloud cover and processed using the same masking logic as in the script). Users then define training points interactively by selecting one of the pre-labelled class names and clicking on the map to place points (one click per point). Each click stores the coordinates and class index in a local point array. Once points are added, users trigger the runs of the machine learning algorithm, which samples spectral values from the composite and trains a Random Forest model, in everything identical to the one in the scripted version. In cases where desired land cover classes are not available in the predefined list of labels, users can assign generic placeholder labels (class1 through class20) from the dropdown menu. These temporary class names can be manually reclassified into meaningful categories later using external GIS software. After the classification, the output map is displayed using a multi-class color palette with an automatically generated legend based on the classes used in the training dataset. To export the results, users define a region by clicking a point at the center of their AOI. The classified raster is clipped to the bounding box that encloses a circular area with a 40 km radius centered on the selected point. The resulting image is then made available for download via a direct export URL.

2.2.2. R Shiny Apps

The R Shiny applications included in the toolkit are designed to visualize godwit telemetry data in relation to habitat conditions in multiple key sites along the flyway, derived from the remote sensing layers produced through GEE workflows. Each app combines user interfaces with spatial filtering, interactive mapping, and summary statistics. These tools are intended for conservation professionals, including managers and policy makers, who can benefit from retrieving quick, descriptive figures on godwit movement in relation to habitat characteristics, and interpret up-to-date godwit responses to changes in their environment. The apps load preprocessed telemetry datasets alongside raster layers that are generated

using the workflows described in section 2.2.1. These inputs are temporally and spatially aligned, and the outputs, including interactive maps of godwit locations and trajectories overlaid on environmental layers, are dynamically updated through automated workflows that are triggered by user-defined filters such as date range, AOI, or individual ID.

Flyway Movement Viewer

The Flyway Movement Viewer provides an interactive overview of godwit movements and the availability of tracking data across the East Atlantic Flyway. Users can define an AOI by selecting a country, uploading a spatial file (either .shp, .geojson, .gpkg, or .kml), or drawing a polygon directly on the map. Additional filters allow users to select a two-year date range, enabling continuous visualization of a migration year, which contains events that span across calendar years. Users can also select a specific date range within the selected migration year, and optionally filter by individual bird, or tagging site, which indicates the country where a bird was originally equipped with a tracking device.

Godwit locations are visualized as interactive map features, with trajectories displayed when a single individual is selected. A time-series plot of latitude, illustrating the migratory timing for specific or multiple birds, is automatically computed from the user-defined filters, as well as a summary panel that reports the number of individuals and the total location fixes within the current filtering settings. When a single bird is selected, its track is segmented into discrete visits to the defined AOI. A visit is defined as a continuous sequence of location fixes within the AOI, where entry is marked by the first within-AOI fix following a location outside the AOI, and exit is inferred from the first subsequent out-of-AOI fix. The segmentation is performed using run-length encoding, which labels AOI over time for each bird. Visit duration is estimated by calculating the time difference between the first and last fix within each visit. These visit summaries are updated dynamically based on user inputs such as date range, AOI geometry, and bird ID selection. No fixed threshold is applied for minimum visit length or number of fixes, meaning that even brief entries into the AOI are retained. Conversely, long staging events that are interrupted by brief excursions of a bird outside the AOI will be segmented in multiple 'visits', while analysts will typically require the full data.

Grassland Production intensity Viewer in Southwest Friesland

The Friesland GPI Viewer visualizes godwit breeding season locations in relation to an annual GPI raster. The app focuses on the southwest Friesland region during the breeding period (April – July). Users can filter location data by year and select individual birds for detailed exploration. Movement points are displayed over a GPI raster from April corresponding to the selected year. If an individual is selected, its full trajectory over the breeding period is displayed. The app uses the *terra* package (Hijmans 2025) in R to annotate each telemetry point with the underlying GPI raster value. GPI values are scaled from 0 to 1 using min-max normalization to facilitate interpretation. After the user selection of filters, a histogram is computed showing the number of godwit locations by GPI value, updating dynamically as filters are applied.

Doñana Wetland Seasonal Flooding Viewer

The Doñana Wetland Seasonal Flooding Viewer links godwit movement data with surface water availability (indicating the flooding status of each pixel) in the Doñana wetlands, a key non-breeding stopover site in southern Spain. The app overlays bird locations on monthly SWIR1-based composite rasters that serve as proxies for surface inundation (see Section

2.1.2). Users select a year and month, which triggers the display of the corresponding surface water raster. Each godwit location is then classified as “flooded” or “dry” based on the underlying SWIR1 pixel value, using a fixed reflectance threshold of 0.186 (as described by Diaz-Delgado et al., 2016). For the selected year and month, the boxplot compares SWIR values at used bird locations with values from 10 randomly sampled available locations across the study area. The dashed horizontal line at 0.186 marks the flooding threshold. Alongside the boxplot, a barplot shows the proportion of GPS locations falling in flooded versus non-flooded pixels, with each bar annotated by the corresponding number of locations.

Senegal Delta Habitat Use Viewer

The Senegal Delta Habitat Use Viewer allows users to visually explore how godwits use different land cover types throughout the non-breeding period in the Senegal River Delta, by integrating godwit location data with a land cover classification raster produced by the GEE land cover workflow described in Section 2.1.2, and in Craft et al. (2025). Users can filter the dataset by non-breeding residency period (e.g., 2023–2024 for the period spanning godwit presence in the region from July 2023 through February 2024) and by climate season: wet (July–November), dry (December–June), or the full year. Godwit locations are displayed on the map, and when a single individual is selected, its movement path is shown as a trajectory. Each location is linked to a land cover class using the *terra* package (Hijmans, 2025), which is used to produce a bar chart showing the proportion of locations falling within each land cover type, updated in real time with data filter changes. When multiple birds are selected, the plot reflects the aggregated habitat use across individuals.

3. Results

Here we present a step-by-step tutorial for each application, showcasing the interface prompts allowing users to trigger the previously introduced workflows.

3.1. Google Earth Engine apps

3.1.1. Grassland Production Intensity Tool

Step 1: Select AOI

Click the placemaker to drop a point; it becomes the center of a 20 km buffer used to filter nearby Sentinel-2 imagery.



Step 2: Choose date

Select a year and month from the left panel; the app filters Sentinel-2 scenes for that AOI and date range.

1.) Add a placemaker on AOI

2.) Select Year:

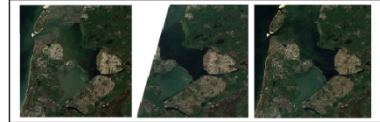
Choose Year

3.) Select Month:

Choose Month

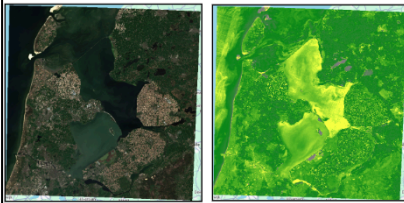
Step 3: Review available images

The app shows the three least-cloudy Sentinel-2 scenes from that month as RGB thumbnails.



Step 4: Select an image for analysis

Click "Select Image" to load the selected scene with a GPI overlay (yellow = low, green = high) for quick comparison of the imagery and GPI.



Step 5: Draw a boundary

Draw a polygon to set the export boundary within your study area.



Step 6: Download the GPI image

After setting the export region, the app provides a link to download the clipped GPI raster as a 20 m GeoTIFF.

[Download Clipped Image](#)

Figure 2. Step-by-step workflow of the Grassland Production Intensity (GPI) GEE App.

3.1.2. Seasonal Surface Water Mapping Tool

Step 1: Select AOI

Click on the map to set a point; the app creates a 30 km buffer AOI for filtering and processing.



Step 2: Choose date

Select a year and month in the left panel; the app builds a median composite from available Landsat 7–9 imagery.

Year:

Month:

Step 3: View the SWIR image

The composite appears on the map; darker blues show wetter or flooded areas.



Step 4: Export the image

Click Export Image to get a download link for the SWIR1 median GeoTIFF, clipped to the 30 km AOI at 30 m resolution.

[Export Image](#)

[Download your image here](#)

Figure 3. Step-by-step workflow of the Seasonal Surface Water Mapping GEE App.

3.1.3. Land Cover Classification Tool

Step 1: Select date

Choose a year (and optional month); if no month is set, an annual median composite is created.

Select Year:
2025 ▾
Select Month:
All Months ▾

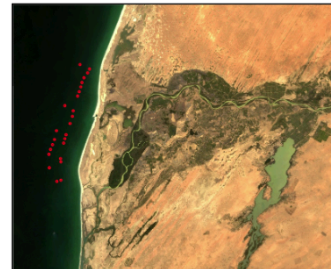
Step 2: Choose land cover class

Select a land cover class from the dropdown.

Open Water
Estuarine Open Water
Brackish Ponds
Aquaculture Ponds
Salt Production Ponds
Freshwater Marsh
Seasonal Wetlands
Floodplain Wetlands

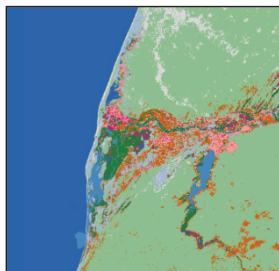
Step 3: Add training points

Click on the map to add training points for the chosen class. Switch classes anytime; use 'Clear Points' to reset.



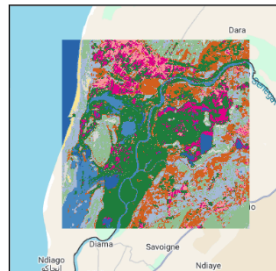
Step 4: Run the classification

Click 'Classify Map'; the app builds a Landsat composite and displays a color-coded land cover map.



Step 5: Define export area

Click 'Select Export Area', then click the map; the app creates a 20 km buffer, clips the classification, and shows the clipped map.



Step 6: Download image

The app gives a download link for the clipped classification as a 30 m GeoTIFF.

Download your clipped image:
<https://earthengine.googleapis.com/v1/projects/legacy/thumbnails/5517e35fe979d7905fa4ce883885e3c3da751ab5b662063db7cc5:getF>

Figure 4. Step-by-step workflow of the Land Cover Classification GEE App.

3.2. R Shiny apps

3.2.1. Flyway Movement Viewer

Step 1: Define AOI

Choose a country, draw a polygon, or upload a spatial file. The selected AOI is highlighted on the map and used to filter movement data.

Select pre-defined aoi (country) or custom

Custom

Upload custom aoi (shp/geojson/gpkg/kml)

Browse... No file selected

Step 2: Choose date

Select a migration year from the dropdown, then refine the time window with the date slider to filter telemetry data.

Select year

2024-2025

Select date range

2024-01-01 2024-02-26 2024-04-22 2024-06-1

Step 3: View godwit GPS points

Filtered godwit locations appear as points on the map. If a single bird is chosen, its full trajectory is shown as a connected line.

Step 4: View latitude-time plot

A time-series plot below the map shows latitude over time; view all birds or a single individual to compare north-south movements.

Step 5: View summary statistics

A summary table shows key metrics: individuals in the AOI, fixes, sex ratio, and proportion of godwits within the site.

metric	value
individuals in aoi	10
fixes in aoi	1191
percent individuals in aoi (%)	5.9
sex ratio (m:f)	M:2 F:3

Step 6: Download summary

Download a table for detailed visit summaries, with start/end dates, duration, and number of fixes per visit, as well as link to the Movebank study page for contacting data owners.

trackid	visits	total_fixes	visit_start	visit_end
Rosalie	1	3	26/02/2025	26/02/2025
Marcellina	1	3	06/03/2024	06/03/2024
Einde.Lotek	2	51	07/01/2024	22/03/2024

Figure 5. Step-by-step workflow of the Flyway Movement Viewer R Shiny App.

3.2.2. Grassland Production Intensity Viewer in Southwest Friesland

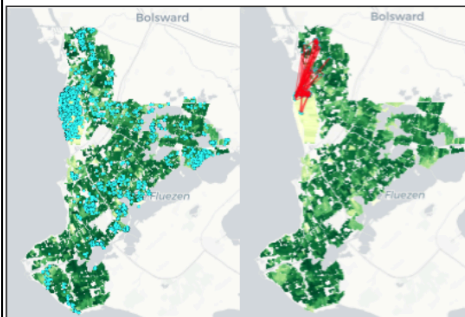
Step 1: Select a year

Select a year from the dropdown to load that year's GPI raster for southwest Friesland.

Select Year
2024

Step 3: View locations and GPI

After selecting a year, the GPI raster appears with GPS points overlaid. If a bird is chosen, its full seasonal path is shown as a connected line.



Step 2: Select birds

Optionally select a bird from the dropdown; otherwise, all GPS points overlapping the year's GPI raster are shown.

select individual
All Birds

Step 4: View GPI plots

A histogram shows the distribution of GPI values, while boxplots compare values across individuals. Both update automatically with year or bird changes.

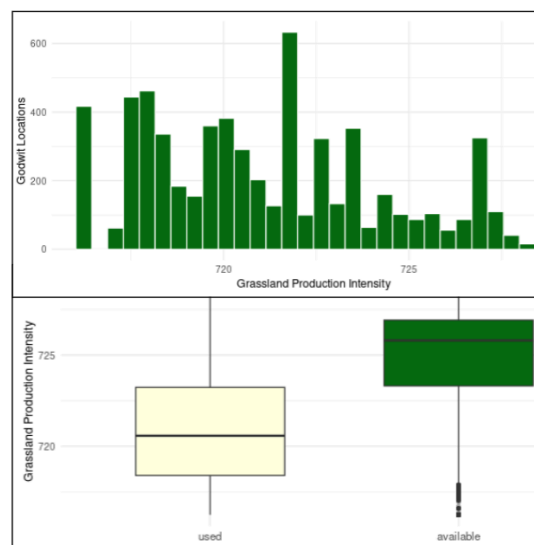


Figure 6. Step-by-step workflow of the Grassland Production Intensity (GPI) R Shiny App.

3.2.3. Doñana Wetland Seasonal Flooding Viewer

Step 1: Select date

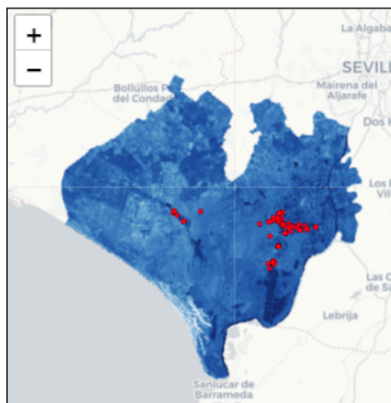
Select a year and month from the panel to load the SWIR1 composite, built from monthly Landsat 7–9 data.

Select Year
2025

Select Month
January

Step 2: View locations

The selected SWIR1 composite appears in blue–grey (darker blue = wetter), with godwit GPS points for that month overlaid.



Step 3: View summary statistics

Below the map, a boxplot compares flooding at used vs. available sites, and a barplot shows GPS fixes in flooded vs. dry pixels, with counts above each bar.

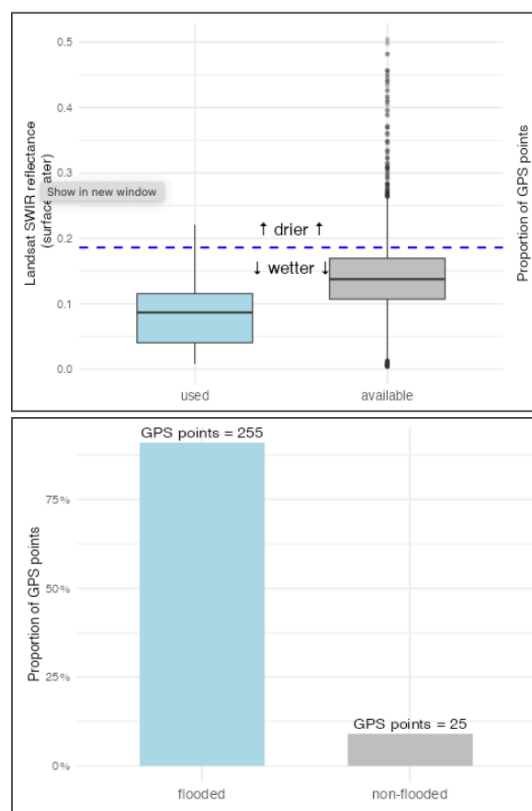


Figure 7. Step-by-step workflow of the Doñana Wetland Seasonal Flooding R Shiny App.

3.2.4. Senegal Delta Habitat Use Viewer

Step 1: Select a migration year and season

Choose a migration year and season (wet, dry, or full year) from the sidebar to load matching GPS locations.

Select Year

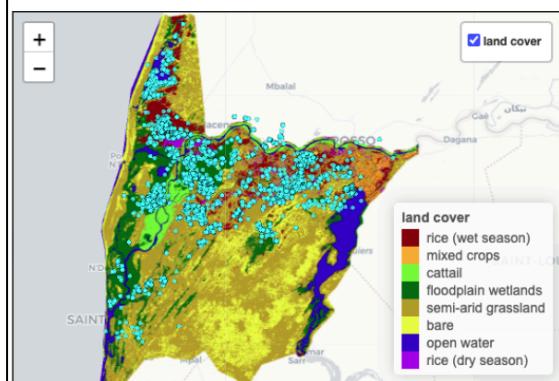
2024-2025

Select Season

All

Step 2: View map and godwit locations

After selecting a season, the 2022 land cover map for the Senegal Delta loads with filtered GPS points.



Step 3: View habitat use summary

A bar chart shows the share of GPS fixes in each land cover type, updating with year, season, and bird selection. For multiple birds, it aggregates habitat use.

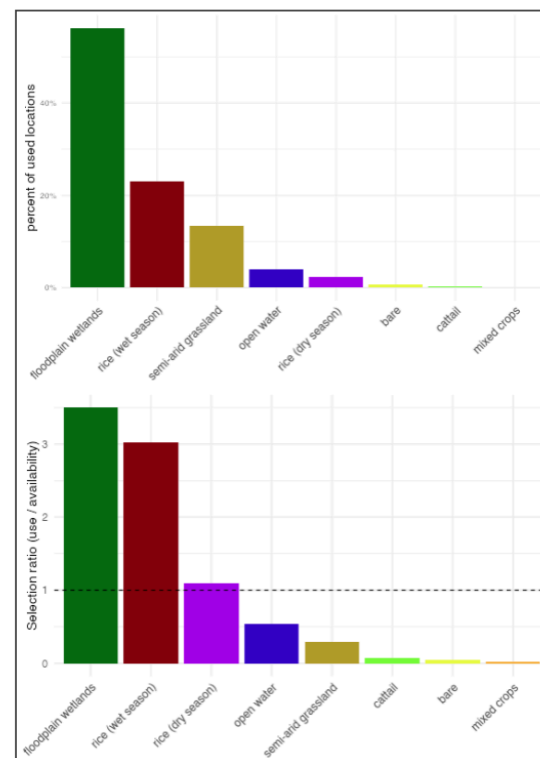


Figure 8. Step-by-step workflow of the Senegal Delta Habitat Use Viewer R Shiny App.

4. Discussion

Despite the rapid growth of ecological remote sensing and animal movement data, much remains to be done to connect data availability and practical conservation use. Many applied ecologists and conservation professionals face urgent questions about habitat loss, migration shifts, or site use that existing datasets could answer, and yet, the technical demands of handling these data often exceed available time, expertise, or infrastructure. This disconnect slows the translation of ecological knowledge into actionable decisions, particularly in systems like that of migratory meadow birds, for which the black-tailed godwit constitutes a flagship species, where effective conservation strongly interacts with human agricultural activities, and depends on understanding space and habitat use across spatial-temporal scales. The toolkit presented here deals with this accessibility and disconnection issue directly by filtering large, complex ecological data through intuitive interfaces that are openly available. It makes site-level assessments (e.g., flooded vs. dry status of a wetland) possible, as well as broader-scale visualizations of movement and connectivity, using the same tools. This accessibility is particularly valuable in interdisciplinary collaborations, where multiple stakeholders may need to work from a shared spatial understanding of animal–habitat interactions, but bring different types of expertise.

Although the toolkit is developed around godwits, its structure is not species-specific. Any migratory system with basic tracking and environmental data could adopt or adapt this approach. Additionally, the modularity of the apps also lends itself to flexible repurposing. For instance, tools currently used to monitor stopover conditions in Doñana could be adapted to examine drought risk in nearby Mediterranean wetlands (Ramirez et al., 2018), or even applied to evaluate water management in rice paddies further away in East Asia (Choi et al., 2022). Similarly, the land cover classification app can be employed to map land cover virtually anywhere in the world without needing any adjustments to the current form. As such, the toolkit offers a generalizable workflow that conservation professionals and ecological researchers can apply to many of the questions surrounding mobile species in changing landscapes.

Future developments could include expansion of geographic coverage by incorporating other godwit subspecies, *L. l. islandica*, *L. l. bohaili*, and *L. l. melanuroides*, which would extend the toolkit's applicability to regions already generating valuable godwit habitat-linked movement data (Nightingale et al., 2023; Zhu et al., 2023; Das et al., 2025). Integrating these systems would support cross-flyway comparisons and contribute to more globally coordinated conservation efforts (Navedo & Piersma, 2023). Additionally, we aim to support new users by organizing onboarding workshops in collaboration with regional conservation partners, as well as offering multilingual interfaces and documentation. The goal is not only broader adoption but also building capacity for users to adapt and extend the toolkit in ways that meet evolving conservation needs. These tools, along with the broader effort to foster their uptake, contribute to open science (Miedema, 2022) by advancing at least three of its key dimensions (Fecher and Frieseke, 2014): technological infrastructure, accessible knowledge production, and collaborative research.

Rather than presenting a finished product, the toolkit is intended as part of an evolving process shaped by co-learning from user experience. While we do not claim that it delivers inclusive decision-making on its own, it helps enable such processes by lowering the technical barriers that often limit participation. The ability to draw a custom AOI and download environmental data through a browser without writing any code may seem minor from a technical standpoint, but from a capacity and access perspective, it is certainly not. In conservation contexts where digital exclusion remains a real constraint, this kind of accessibility is essential. Our aim is not to overstate the toolkit's potential impact, but to

frame it as one step towards building a more participatory and equitable knowledge infrastructure – moving beyond simply providing access to data, towards supporting practices where data become usable, situated knowledge.

5. Data availability

All applications and source code associated with the Godwit Landscapes Toolkit are publicly accessible through dedicated platforms. The toolkit includes both GEE scripts for on-demand environmental data processing and R Shiny applications for interactive exploration of godwit movement and habitat use.

5.1. Google Earth Engine applications

Browser-based apps:

1. Grassland Production Intensity Tool:
<https://ee-tbcraft.projects.earthengine.app/view/grasslandproductionintensity>
2. Seasonal Surface Water Mapping Tool:
<https://ee-tbcraft.projects.earthengine.app/view/floodmapping>
3. Land Cover Classification Tool:
<https://ee-tbcraft.projects.earthengine.app/view/landcoverclassificationapp>

Corresponding code editor scripts (requires GEE account):

1. Grassland Production Intensity (GPI) Tool Script:
<https://code.earthengine.google.com/a0eb88e3af93be5d39d911b2f4b18bf3>
2. Seasonal Surface Water Mapping Tool Script:
<https://code.earthengine.google.com/e24a603dabcf0c22e4431e72d2522af6>
3. Land Cover Classification Tool Script:
<https://code.earthengine.google.com/a7832e9efc471371b71fed6546b72413>

5.2. R Shiny applications

1. Flyway Movement Viewer:
https://tbcraft.shinyapps.io/flyway_tracking_summary_app/
2. Grassland Production intensity Viewer in Southwest Friesland:
https://tbcraft.shinyapps.io/Friesland_GPI_App/
3. Doñana Wetland Seasonal Flooding Viewer:
https://tbcraft.shinyapps.io/Donana_Wetland_Viewer/
4. Senegal Delta Habitat Use Viewer:
https://tbcraft.shinyapps.io/Senegal_Delta_Habitat_Use_App/

5.3. Code repository

All app source code, data processing scripts, and deployment workflows are available in the GitHub repository: https://github.com/taylorbcraft/Godwit_toolkit

An archived version of the toolkit is also available on Zenodo:

<https://doi.org/10.5281/zenodo.15628718>

5.4. Tracking data projects

The telemetry data used in the Godwit Landscapes toolkit originate from long-term collaborative tagging and funding efforts among migratory waterbird research and conservation groups, distributed across Europe, including the Netherlands, Germany, Portugal, and Spain. All telemetry data are stored across 36 Movebank studies (see Table 2 for Movebank study names and associated project or funding sources), and the collection is continuously updated as new studies are added each year. At present, the dataset includes telemetry from over 500 godwits tagged with Argos PTT and GPS devices, resulting in over one million recorded locations since 2013.

Table 2. Overview of Movebank studies and their associated projects or funding sources contributing godwit telemetry data to the Godwit Landscapes toolkit.

Movebank study name	Project/Funding
Black-tailed Godwit Iberia - NL 2019	ALW-NWO TOP grant "Shorebirds in Space"; The Province of Fryslan; NWO Spinoza-Prize
Black-tailed Godwit NL 2021 Microwave	Ministry of Agriculture, Fisheries, Food Security and Nature (LNVN)
Black-tailed Godwits -- VeenVitaal	Veen Vitaal
Black-tailed Godwits Extremadura (HQXS)	MCIN/AEI/10.13039/501100011033
Black-tailed Godwits Extremadura (Lotek)	MCIN/AEI/10.13039/501100011033
Black-tailed godwits NL South Holland	Province Zuid-holland
BtgTagus2021	FCT/MCTES
Duemmer Adult Godwits 2018	Lower Saxon State Department for Waterway, Coastal and Nature Conservation (NLWKN)
Duemmer Adult Godwits 2019	Lower Saxon State Department for Waterway, Coastal and Nature Conservation (NLWKN)
Duemmer Adult Godwits 2020	Lower Saxon State Department for Waterway, Coastal and Nature Conservation (NLWKN)
Duemmer Adult Godwits 2021	LIFE IP GrassBirdHabitats
Duemmer Godwit Chicks 2018	LIFE IP GrassBirdHabitats
Duemmer Godwit Chicks 2019	LIFE IP GrassBirdHabitats
Duemmer Godwit Chicks 2020	LIFE IP GrassBirdHabitats

Duemmer Godwit Chicks 2021	LIFE IP GrassBirdHabitats
GrassBirdHabitats Godwit Chicks 2022	LIFE IP GrassBirdHabitats
GrassBirdHabitats Godwit Chicks 2023	LIFE IP GrassBirdHabitats
GrassBirdHabitats Godwit Chicks 2024	LIFE IP GrassBirdHabitats
Haanmeer Adults	ALW-NWO TOP grant "Shorebirds in Space"; The Province of Fryslan; NWO Spinoza-Prize
Hand-raised Juveniles	ALW-NWO TOP grant "Shorebirds in Space"; The Province of Fryslan; NWO Spinoza-Prize
hand-raised juveniles 2016	ALW-NWO TOP grant "Shorebirds in Space"; The Province of Fryslan; NWO Spinoza-Prize
hand-raised juveniles 2017	ALW-NWO TOP grant "Shorebirds in Space"; The Province of Fryslan; NWO Spinoza-Prize
HQXS Black-tailed Godwits	MCIN/AEI/10.13039/501100011033
Iberia Blackwits	ALW-NWO TOP grant "Shorebirds in Space"; The Province of Fryslan; NWO Spinoza-Prize
ICARUS Black-Tailed Godwits Theunis Piersma	Ministry of Agriculture, Fisheries, Food Security and Nature (LVVN)
Polish Adults	ALW-NWO TOP grant "Shorebirds in Space"; The Province of Fryslan; NWO Spinoza-Prize
Teampiersma Global Messenger 2025	Ministry of Agriculture, Fisheries, Food Security and Nature (LVVN)
Teampiersma HQXS 2021	Ministry of Agriculture, Fisheries, Food Security and Nature (LVVN)
Teampiersma HQXS 2022	Ministry of Agriculture, Fisheries, Food Security and Nature (LVVN)
Teampiersma HQXS 2023	Ministry of Agriculture, Fisheries, Food Security and Nature (LVVN)
Teampiersma HQXS 2024	Ministry of Agriculture, Fisheries, Food Security and Nature (LVVN)
Teampiersma Interrex 2023	Ministry of Agriculture, Fisheries, Food Security and Nature (LVVN)
Teampiersma Interrex Druid 2025	Ministry of Agriculture, Fisheries, Food Security and Nature (LVVN)
Wild Juveniles	ALW-NWO TOP grant "Shorebirds in Space"; The Province of Fryslan; NWO Spinoza-Prize
Wild Juveniles 2	ALW-NWO TOP grant "Shorebirds in Space"; The Province of Fryslan; NWO Spinoza-Prize
Wild Juveniles 3	ALW-NWO TOP grant "Shorebirds in Space"; The Province of Fryslan; NWO Spinoza-Prize

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